

Full Stack Assignment-1

A.Y 2025-26

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Q1.

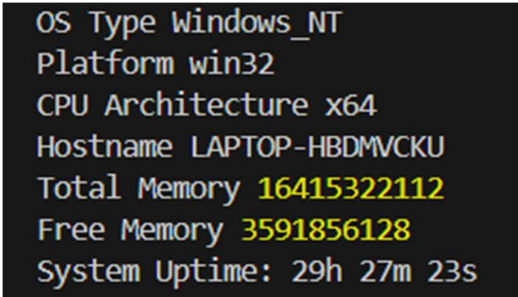
```
import os from 'os';

console.log("OS Type", os.type());
console.log("Platform", os.platform());
console.log("CPU Architecture", os.arch());
console.log("Hostname", os.hostname());
console.log("Total Memory", os.totalmem());
console.log("Free Memory", os.freemem());

let uptimeSeconds= os.uptime();

const hours = Math.floor(uptimeSeconds / 3600);
const minutes = Math.floor((uptimeSeconds % 3600) / 60);
const seconds = Math.floor(uptimeSeconds % 60);

console.log("System Uptime: " + hours + "h " + minutes + "m " + seconds + "s");
```

A terminal window with a black background and light blue text. It displays the output of the JavaScript code from the previous block. The output shows system information: OS Type Windows_NT, Platform win32, CPU Architecture x64, Hostname LAPTOP-HBDMVCKU, Total Memory 16415322112, Free Memory 3591856128, and System Uptime: 29h 27m 23s.

```
OS Type Windows_NT
Platform win32
CPU Architecture x64
Hostname LAPTOP-HBDMVCKU
Total Memory 16415322112
Free Memory 3591856128
System Uptime: 29h 27m 23s
```

Q2.

```
import path from 'path';

import { fileURLToPath } from 'url';
```

```

const __filename = fileURLToPath(import.meta.url);
const __dirname = path.dirname(__filename);
console.log("Current File Name:", path.basename(__filename));
console.log("Current Directory Name:", __dirname);
console.log("File Extension:", path.extname(__filename));
const newFilePath = path.join(__dirname, 'folder', 'example.txt');
console.log("New File Path using path.join():", newFilePath);
const normalizedPath = path.normalize('/users/demo/./docs/file.txt');
console.log("Normalized Path:", normalizedPath);
const absolutePath = path.resolve('data', 'info.txt');
console.log("Absolute Path using path.resolve():", absolutePath);

```

```

Current File Name: demo.js
Current Directory Name: C:\Users\Administrator\Desktop\Full Stack Ass\ass-1
File Extension: .js
New File Path using path.join(): C:\Users\Administrator\Desktop\Full Stack Ass\ass-1\folder\example.txt
Normalized Path: \users\docs\file.txt
Absolute Path using path.resolve(): C:\Users\Administrator\Desktop\Full Stack Ass\ass-1\data\info.txt

```

Q3.

```

import colors from 'colors';

console.log(colors.green.bgWhite("Welcome to IMCC"));

console.log("\n==== Custom Theme Demonstration =====\n");

colors.setTheme({

  value: 'cyan',

  success: 'green',

  warning: 'magenta',

  error: ['red', 'bold'],

  note: ['white', 'underline'],

  title: ['white', 'bgBlue']

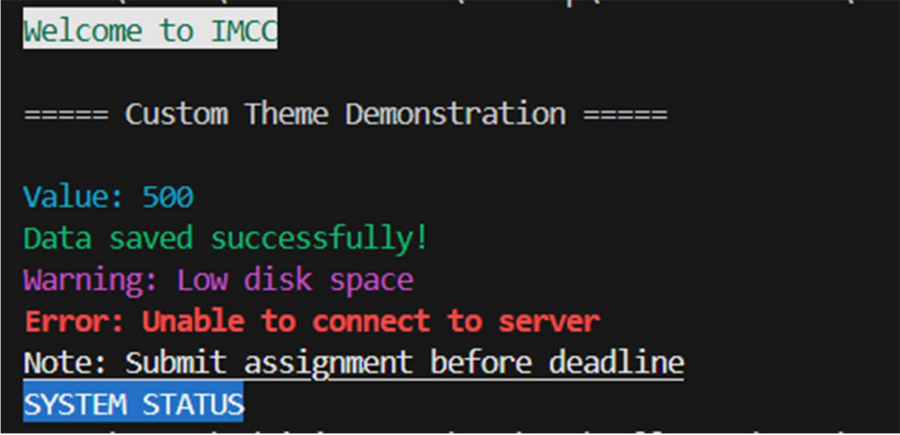
});

```

```

console.log("Value: 500".value);
console.log("Data saved successfully!".success);
console.log("Warning: Low disk space".warning);
console.log("Error: Unable to connect to server".error);
console.log("Note: Submit assignment before deadline".note);
console.log("SYSTEM STATUS".title);

```



```

Welcome to IMCC

==== Custom Theme Demonstration =====

Value: 500
Data saved successfully!
Warning: Low disk space
Error: Unable to connect to server
Note: Submit assignment before deadline
SYSTEM STATUS

```

Q4.

```

import fs from 'fs';

try {

  const filePath = 'myfile.txt';

  if (!fs.existsSync(filePath)) {

    fs.writeFileSync(filePath, "Welcome to IMCC!\nFull Stack Development 2026.");

  }

  const data = fs.readFileSync(filePath, 'utf8');

  const characters = data.length;

  const words = data.trim().split(/\s+/).length;

  const lines = data.split("\n").length;

  const vowels = (data.match(/[aeiou]/gi) || []).length;

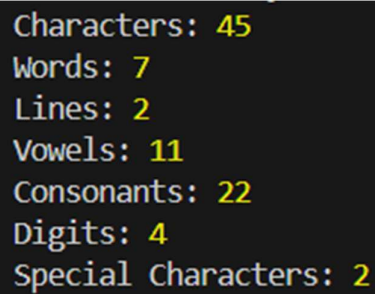
  const consonants = (data.match(/[bcdfghjklmnpqrstvwxyz]/gi) || []).length;

```

```

const digits = (data.match(/[0-9]/g) || []).length;
const specialChars = (data.match(/[^a-zA-Z0-9\s]/g) || []).length;
console.log("Characters:", characters);
console.log("Words:", words);
console.log("Lines:", lines);
console.log("Vowels:", vowels);
console.log("Consonants:", consonants);
console.log("Digits:", digits);
console.log("Special Characters:", specialChars);
} catch (err) {
  console.error("Error:", err.message);
}

```



```

Characters: 45
Words: 7
Lines: 2
Vowels: 11
Consonants: 22
Digits: 4
Special Characters: 2

```

Q5.

```

import XLSX from 'xlsx';

try {
  const studentData = [
    { Name: "Amit", Age: 20, Weight: 65 },
    { Name: "Riya", Age: 22, Weight: 55 }
  ];

  const ws = XLSX.utils.json_to_sheet(studentData);
  const wb = XLSX.utils.book_new();
  XLSX.utils.book_append_sheet(wb, ws, "Students");
}

```

```

XLSX.writeFile(wb, "student.xlsx");

console.log("student.xlsx created.");

const readWb = XLSX.readFile("student.xlsx");

const readWs = readWb.Sheets["Students"];

const jsonData = XLSX.utils.sheet_to_json(readWs);

console.log("Excel Data in JSON:");

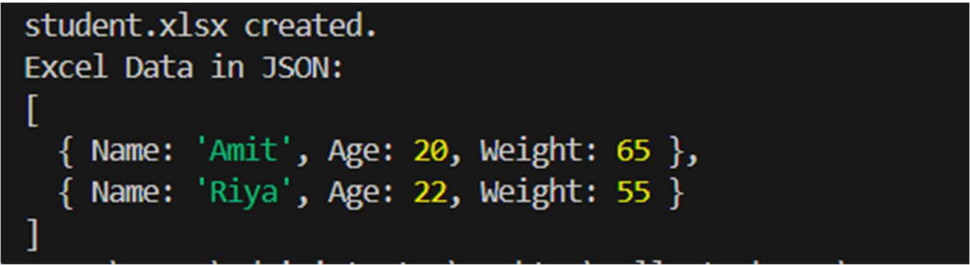
console.log(jsonData);

} catch (err) {

  console.error("Error:", err.message);

}

```



```

student.xlsx created.
Excel Data in JSON:
[
  { Name: 'Amit', Age: 20, Weight: 65 },
  { Name: 'Riya', Age: 22, Weight: 55 }
]

```

Q6.

```

import fs from 'fs';

try {

  const empData = [

    { Name: "Raj", Email: "raj@gmail.com", JobProfile: "Developer" },

    { Name: "Sneha", Email: "sneha@gmail.com", JobProfile: "Designer" }

  ];

  fs.writeFileSync('empdata.json', JSON.stringify(empData, null, 2));

  console.log("empdata.json created.");

  const data = fs.readFileSync('empdata.json');

  const parsedData = JSON.parse(data);

  console.log("Employee Data:");

```

```

    console.log(parsedData);
  } catch (err) {
    console.error("Error:", err.message);
  }
}

```

```

empdata.json created.
Employee Data:
[
  { Name: 'Raj', Email: 'raj@gmail.com', JobProfile: 'Developer' },
  { Name: 'Sneha', Email: 'sneha@gmail.com', JobProfile: 'Designer' }
]

```

Q7.

```

import http from 'http';

try {
  const server = http.createServer((req, res) => {
    res.writeHead(200, { 'Content-Type': 'text/plain' });
    res.write("Welcome to Node.js HTTP Server!");
    res.end();
  });

  server.listen(3000, () => {
    console.log("Server running at http://localhost:3000");
  });
} catch (err) {
  console.error("Server Error:", err.message);
}

```

```

Server running at http://localhost:3000

```

Q8.

```

import EventEmitter from 'events';

class Order extends EventEmitter {}

```

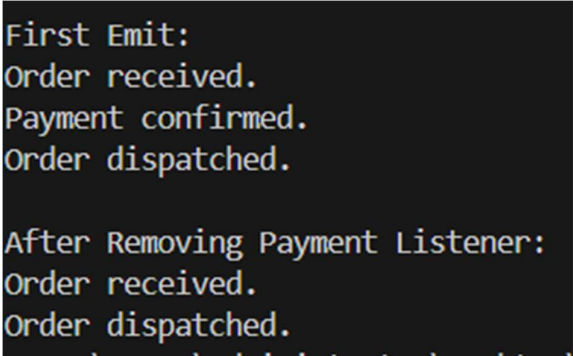
```
const order = new Order();

function orderReceived() {
  console.log("Order received.");
}

function paymentConfirmed() {
  console.log("Payment confirmed.");
}

function orderDispatched() {
  console.log("Order dispatched.");
}

try {
  order.on("orderProcess", orderReceived);
  order.on("orderProcess", paymentConfirmed);
  order.on("orderProcess", orderDispatched);
  console.log("\nFirst Emit:");
  order.emit("orderProcess");
  order.removeListener("orderProcess", paymentConfirmed);
  console.log("\nAfter Removing Payment Listener:");
  order.emit("orderProcess");
} catch (err) {
  console.error("Error:", err.message);
}
```



```
First Emit:
Order received.
Payment confirmed.
Order dispatched.

After Removing Payment Listener:
Order received.
Order dispatched.
```